

Parkinsonism

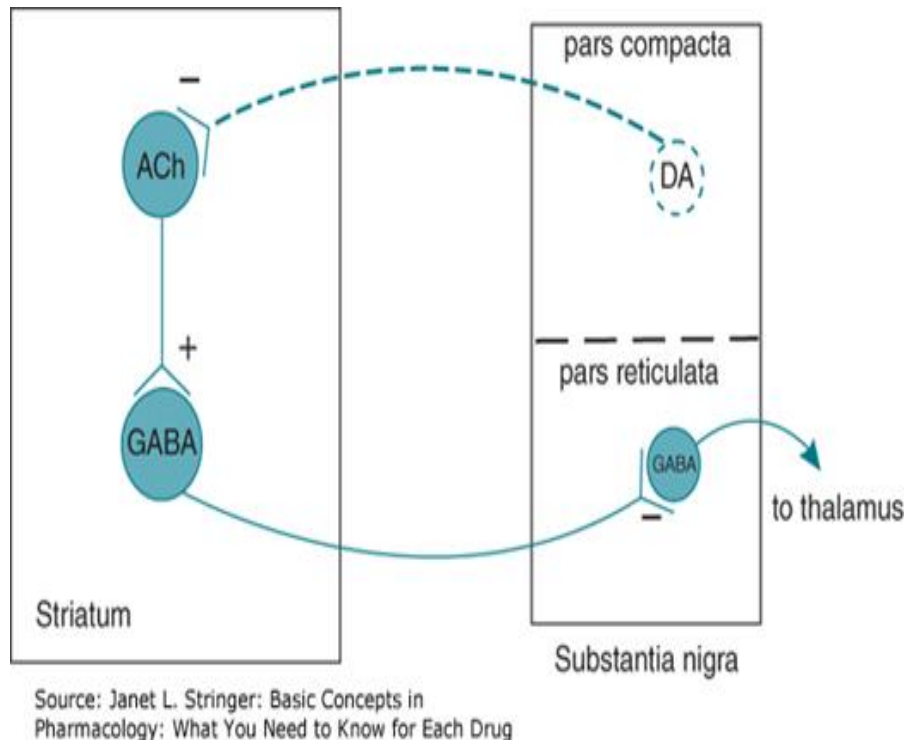
Introduction and pathophysiology

Parkinsonism (paralysis agitans) is a common movement disorder that involves dysfunction in the basal ganglia and associated brain structures.

Signs include rigidity of skeletal muscles, akinesia (or bradykinesia), flat facies, and tremor at rest (mnemonic RAFT).

In advanced parkinsonism, behavioral changes, delusions, and hallucinations may occur.

Pathologic characteristics include a decrease in the levels of striatal dopamine and the degeneration of dopaminergic neurons in the nigrostriatal tract (mostly D2 receptors) that normally inhibit the activity of striatal GABAergic neurons and leading to excessive excitatory actions of cholinergic neurons on striatal GABAergic neurons; thus, dopamine and acetylcholine activities are out of balance in parkinsonism.



Etiology

Natural: genetic predisposition, increasing frequency during aging from the fifth decade of life onward.

Occupational: more incidence in farmers and those with lead or manganese exposure.

Pharmacological: **drug induced.**

Many drugs can cause parkinsonian symptoms; these effects are usually reversible.

The most important precipitating drugs are the butyrophenone and phenothiazine antipsychotic drugs, which block brain dopamine D2 receptors.

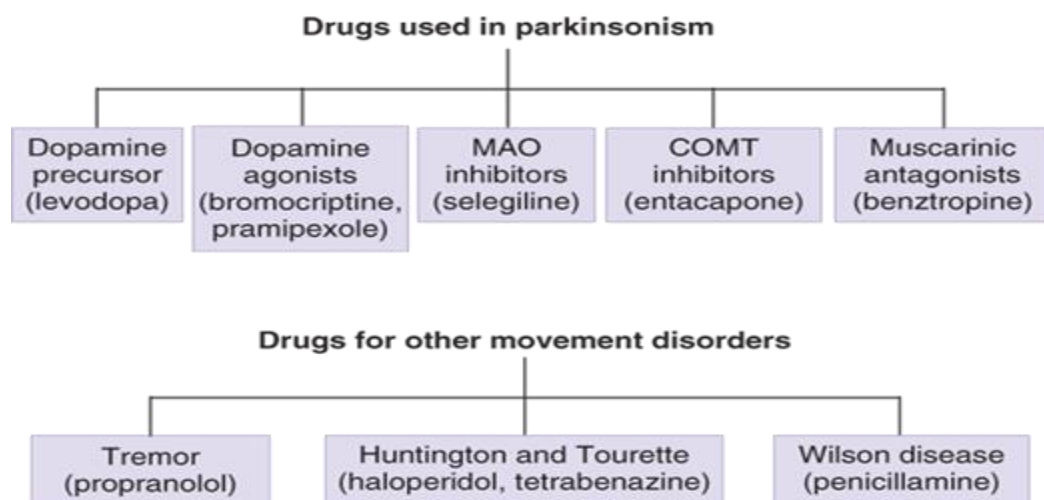
Other drugs include those inhibiting dopamine synthesis eg methyl dopa, storage eg reserpine or causing destruction of dopaminergic neurons eg MPTP (1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine) addiction.

Pharmacotherapy

Many drugs can cause parkinsonian symptoms; these effects are usually reversible.

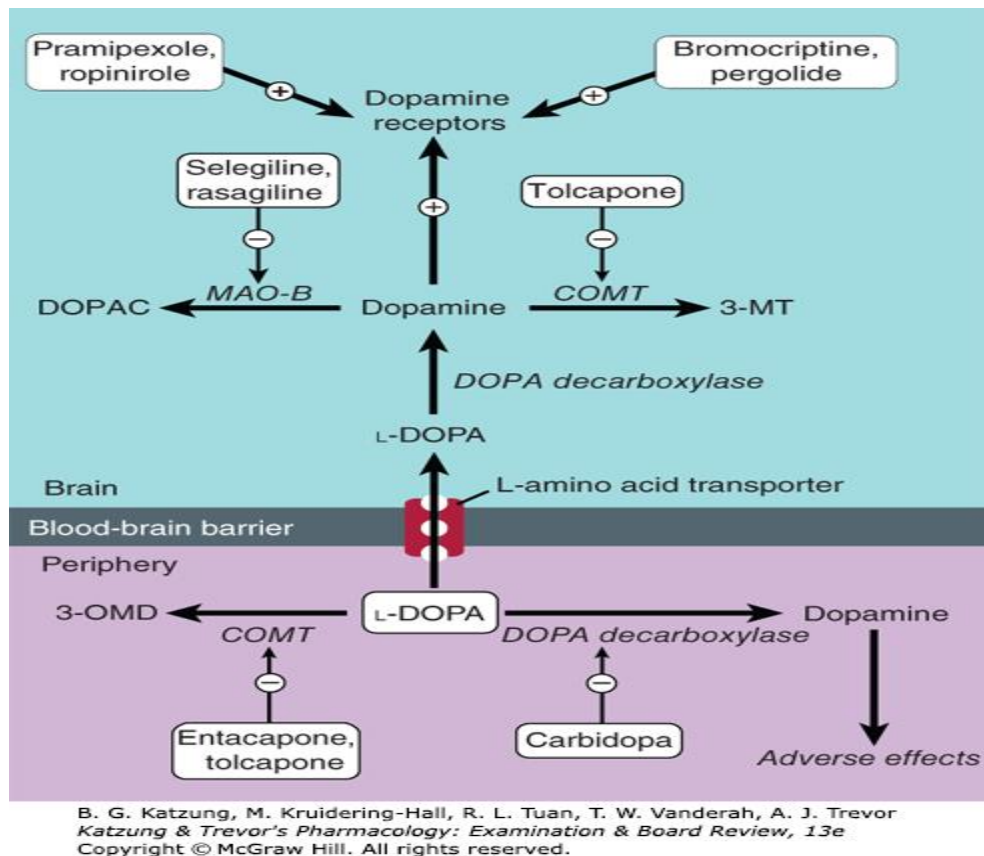
The most important precipitating drugs are the butyrophenone and phenothiazine antipsychotic drugs, which block brain dopamine D2 receptors.

Other drugs include those inhibiting dopamine synthesis eg methyl dopa, storage eg reserpine or causing destruction of dopaminergic neurons eg MPTP (1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine) addiction



Strategies of drug treatment of parkinsonism involve increasing dopamine activity (mostly D2 receptors) in the brain, decreasing muscarinic cholinergic activity in the brain, or both.

Mechanism of action:



A. Levodopa

Because dopamine has low bioavailability and does not readily cross the blood-brain barrier, its precursor, L-dopa (levodopa), is used.

Levodopa is usually given with carbidopa, a drug that does not cross the blood-brain barrier but inhibits dopa decarboxylase in peripheral tissues (used in the U.S.) or benserazide (available outside the U.S.). With this combination, the plasma half-life is prolonged, lower doses of levodopa are effective, and there are fewer peripheral side effects. An extended-release formulation is also available.

Adverse effects:

GIT: nausea, vomiting (tolerance after months), anorexia

CVS: postural hypotension, tachycardia, arrhythmia

CNS: Wearing-off phenomenon (A carbidopa-levodopa extended-release capsule that contains both immediate- and extended-release beads to reduce off time or a levodopa inhalational powder is administered through an inhaler and is also effective for the treatment of off periods) and later on-off phenomenon (Istradefylline, analog of caffeine and a selective antagonist of the adenosine A2A receptor boosting the dopaminergic signaling, may be used for off periods), dyskinesia and behavioral effects such as induction of hallucinations (contraindicated in patients with a history of psychosis) and impulse control disorder.

DDIs: Administration of levodopa with nonspecific inhibitors of MAO accentuates the actions of levodopa and may precipitate life-threatening hypertensive crisis and hyperpyrexia; nonspecific MAO inhibitors always should be discontinued at least 14 days before levodopa is administered (note that this prohibition does not include the MAO-B subtype-specific inhibitors selegiline, rasagiline, and safinamide).

NB. Sudden withdrawal could be fatal; neuroleptic malignant syndrome.

B. Dopamine agonists

1- Pramipexole

Non-ergot drug has high agonist affinity for the dopamine D3 receptor and acts as a scavenger for hydrogen peroxide (neuroprotection).

It is effective as monotherapy in mild parkinsonism and can be used together with levodopa in more advanced disease.

Administered orally 3 times daily and is excreted largely unchanged in the urine.

The dose of pramipexole may require reduction in patients with renal dysfunction.

AEs: anorexia, nausea and vomiting, postural hypotension, and dyskinesias.

Mental disturbances are more common than with levodopa.

CI: active peptic ulcer disease, psychotic illness, or recent myocardial infarction.

2. Ropinirole

Non-ergot drug has high agonist affinity for the dopamine D2 receptor and acts as a scavenger for hydrogen peroxide (neuroprotection).

It is effective as monotherapy in mild parkinsonism and can be used together with levodopa in more advanced disease.

Administered orally 3 times daily but a prolonged release form can be taken once daily.

Metabolized by hepatic CYP1A2, and other drugs metabolized by this isoform (eg, caffeine, warfarin) may reduce its clearance.

AEs and CI: as pramipexole.

3. Rotigotine

A newer dopamine agonist that is administered via a transdermal patch. It is proposed that it provides a slower and more continuous action than the oral dopamine agonists.

4. Bromocriptine

Ergot alkaloid that acts as a partial agonist at dopamine D2 receptors in the brain.

Used as monotherapy, in combinations with levodopa (and with anticholinergic drugs), and in patients who are refractory to or cannot tolerate levodopa.

Common adverse effects include anorexia, nausea and vomiting, dyskinesias, and postural hypotension. Behavioral effects e.g., confusion, hallucinations, occur more commonly with bromocriptine than with newer dopamine agonists.

Ergot-related effects include erythromelalgia and pulmonary infiltrates.

Bromocriptine is now considered obsolete in patients with Parkinson disease.

5. Apomorphine

Injected subcutaneously, may provide rapid (within 10 min) but temporary relief (1–2 h) of off-periods of akinesia when other measures, such as oral DA agonists or COMT inhibitors, have failed to control the off episodes..

AEs: Severe nausea (pretreatment for 3 days with antiemetics other than 5HT₃ antagonist class is necessary), dyskinesias, hypotension, drowsiness, and sweating.

C. Monoamine Oxidase Inhibitors

Selegiline, rasagiline and safinamide are selective inhibitors of monoamine oxidase type B. Selegiline and rasagiline selectively and irreversibly inactivate MAO-B; safinamide is a reversible inhibitor of MAO-B and it also reduces glutamate release .

Selegiline and safinamide has minimal (/no) efficacy in Parkinsonism if given alone but can be used adjunctively with levodopa in early or mild Parkinsonism.

Rasagiline is more potent and has been used as monotherapy in early symptomatic Parkinsonism as well as in combinations with levodopa in advanced cases.

A patient showing wearing OFF for the first-time taking levodopa monotherapy or in combination with a DA agonist is the best candidate for safinamide.

AEs: insomnia, mood changes, dyskinesias, gastrointestinal distress, and hypotension.

Metabolites of selegiline include amphetamine and methamphetamine, which may cause anxiety, insomnia, and other adverse symptoms.

Safinamide dose adjustment is required in moderate hepatic dysfunction, and it should be avoided in severe hepatic disease.

DDIs: Selegiline or rasagiline should not be given in combination with meperidine, tramadol, methadone, propoxyphene dextromethorphan, St. John's wort, and cyclobenzaprine to avoid agitation, delirium, and mortality.

Selegiline has been rarely implicated in the serotonin syndrome when used with serotonin selective reuptake inhibitors or TCAs. Close monitoring and use of low doses of the antidepressant are advisable

D. Catechol-O-methyltransferase Inhibitors

Entacapone (acts only in the periphery), tolcapone and opicapone.

Used as adjuncts to levodopa-carbidopa, decreasing fluctuations (wearing-off), improving response, and prolonging on-time. Tolcapone is taken 3 times daily, entacapone 5 times daily. A formulation combining levodopa, carbidopa, and entacapone is available.

AEs are partly due to increased levels of levodopa include dyskinesias, gastrointestinal distress, and postural hypotension.

L-dopa dose reductions may be needed for the first few days of COMT inhibitor use.

COMT inhibitors may cause sleep disturbances and orange discoloration of the urine.

Only tolcapone increases liver enzymes and has caused acute hepatic failure, necessitating routine monitoring of liver function tests and signed patient consent for use in USA.

Opicapone is a new FDA-approved COMT inhibitor. Despite a short plasma $t_{1/2}$, its slow dissociation rate allows its use once daily. Its use with levodopa reduces off time comparable to entacapone

E. Amantadine

Antiviral drug that may increase synthesis or release of dopamine or inhibit dopamine reuptake as well as muscarinic blocking actions.

May improve bradykinesia, rigidity, and tremor but is usually effective for only a few weeks, yet it's useful as an add-on agent to attenuate dyskinesias.

AEs: Behavioral effects, livedo reticularis, urine retention and postural hypotension.

F. Acetylcholine-Blocking (Antimuscarinic) Drugs

eg, benztropine, biperiden, trihexyphenidyl and orphenadrine

May improve the tremor and rigidity of parkinsonism but have little effect on bradykinesia. Used adjunctively in parkinsonism and also alleviate the reversible extrapyramidal symptoms caused by antipsychotic drugs.

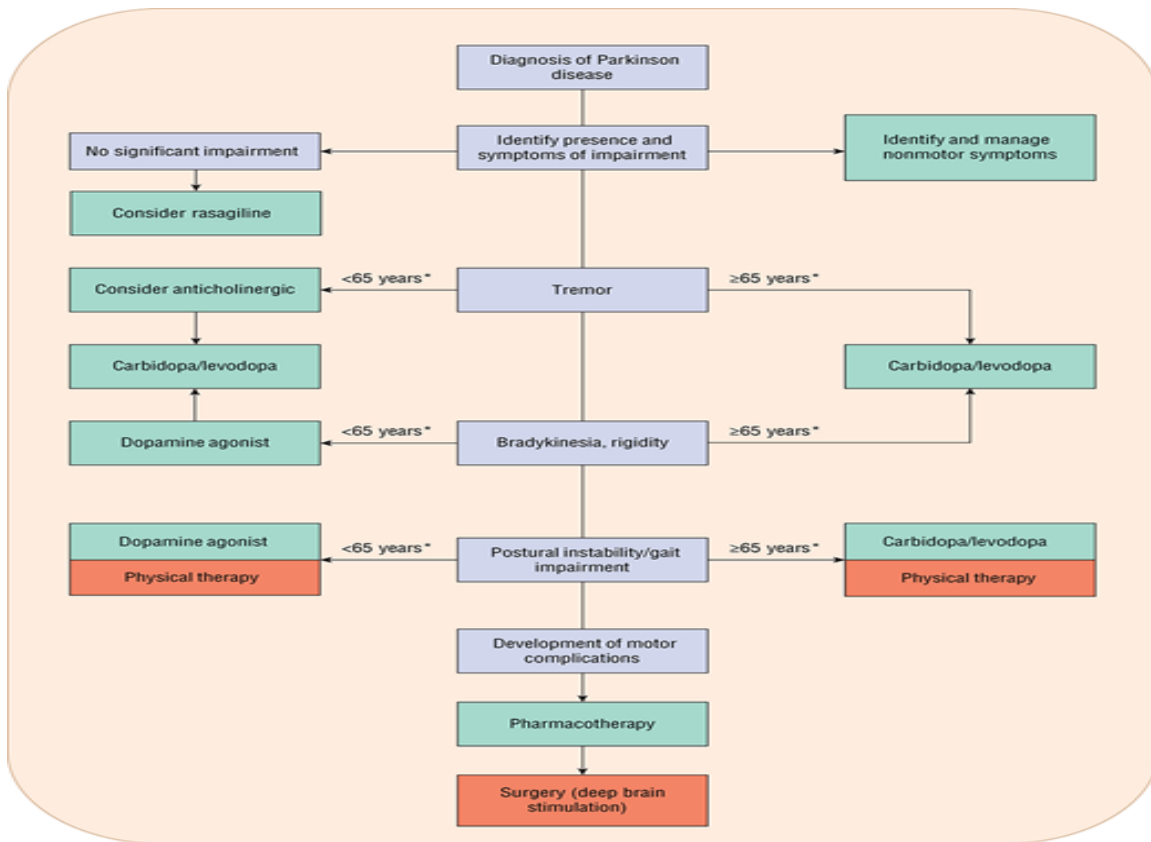
CNS toxicity includes drowsiness, inattention, confusion, delusions, and hallucinations. Peripheral adverse effects are dry mouth, constipation, urinary retention, and confusion.

CI: tardive dyskinesias that result from prolonged use of antipsychotic drugs.

Used with caution in older patients and those with pre-existing cognitive difficulties.

G. Other Drugs

Pimavanserin, a 5-HT_{2A} inverse agonist, is specifically approved for parkinsonism-associated psychosis. Atypical antipsychotics may also be useful.



Source: Joseph T. DiPiro, Gary C. Yee, L. Michael Posey, Stuart T. Haines, Thomas D. Nolin, Vicki L. Ellingrod: *DiPiro: Pharmacotherapy A Pathophysiologic Approach*, 12e Copyright © McGraw Hill. All rights reserved.